

EE331 – Power Electronics Laboratory

Experiment N° 01

Single-Phase Half-Wave / Full-Wave
Rectification Circuit
(Experimental)

Students: Hammachi Nouh Abdelbasset
.....
.....
Group: L3 – G03
Date:
Instructor: Mrs. RECIOUI F.Z.

Contents

1	Objectives	2
2	Equipment	2
3	Pre-lab preparation (theory)	2
3.1	Definitions used in this report	2
3.2	Half-wave rectifier with R load	2
3.3	Half-wave rectifier with R–C load	3
3.4	Full-wave (bridge) rectifier with R load	3
3.5	Full-wave (bridge) rectifier with R–C load	3
3.6	Summary of the predicted values	3
4	Circuits and procedure	4
4.1	Circuit diagrams	4
4.2	Procedure followed	4
5	Results	5
5.1	Measured values	5
5.2	Comparison with theory	5
5.3	Waveforms	5
6	Discussion and answers to questions	9
7	Conclusion	10

1 Objectives

- Build single-phase half-wave and full-wave (bridge) diode rectifiers.
- Observe, on the oscilloscope, the input voltage, the load voltage, the load (diode) current and the diode voltage with a resistive (R) and a resistive-capacitive (R-C) load.
- Measure the DC and AC components of the load voltage and evaluate the performance parameters: RMS value, form factor (F.F.), ripple factor (R.F.) and rectification efficiency (η).
- Compare the experimental results with the theoretical predictions and show the effect of the smoothing capacitor.

2 Equipment

- Trainer module ED-2040-A “Single phase half, full-wave rectifier” (24 V_{rms}, 50 Hz AC source, switch, fuse, diodes D , D_1 – D_4 , 1 Ω current-sensing resistor).
- Load resistor $R_L = 500 \Omega$, filter capacitor $C_L = 1000 \mu\text{F}$.
- Digital oscilloscope with two probes, connecting leads.

3 Pre-lab preparation (theory)

3.1 Definitions used in this report

For a periodic load voltage $v_L(t)$ of period T :

$$V_{dc} = \frac{1}{T} \int_0^T v_L dt, \quad V_{rms} = \sqrt{\frac{1}{T} \int_0^T v_L^2 dt} = \sqrt{V_{dc}^2 + V_{ac}^2},$$
$$\text{F.F.} = \frac{V_{rms}}{V_{dc}}, \quad \text{R.F.} = \frac{V_{ac}}{V_{dc}} = \sqrt{\text{F.F.}^2 - 1}, \quad \eta = \frac{P_{dc}}{P_{ac}} = \frac{V_{dc}^2/R}{V_{rms}^2/R} = \left(\frac{V_{dc}}{V_{rms}} \right)^2.$$

V_{ac} is the RMS value of the AC (ripple) component only. The supply is $v_s(t) = V_m \sin \omega t$ with

$$V_m = \sqrt{2} \times 24 = 33.94 \text{ V}, \quad f = 50 \text{ Hz}, \quad T = 20 \text{ ms}, \quad \tau = R_L C_L = 500 \times 1000 \cdot 10^{-6} = 0.5 \text{ s}.$$

The 1 Ω sensing resistor is negligible compared with R_L ($1 \ll 500$).

3.2 Half-wave rectifier with R load

The diode conducts only when $v_s > 0$: $v_L = V_m \sin \omega t$ for $0 < \omega t < \pi$ and $v_L = 0$ for $\pi < \omega t < 2\pi$.

$$V_{dc} = \frac{1}{2\pi} \int_0^\pi V_m \sin \theta d\theta = \frac{V_m}{\pi} = \frac{33.94}{\pi} = 10.80 \text{ V}$$
$$V_{rms} = \sqrt{\frac{1}{2\pi} \int_0^\pi V_m^2 \sin^2 \theta d\theta} = \frac{V_m}{2} = 16.97 \text{ V}$$
$$V_{ac} = \sqrt{V_{rms}^2 - V_{dc}^2} = 13.09 \text{ V}$$
$$\text{F.F.} = \frac{\pi}{2} = 1.571, \quad \text{R.F.} = \sqrt{1.571^2 - 1} = 1.211, \quad \eta = \frac{4}{\pi^2} = 40.5 \text{ \%}.$$

Peak load current: $I_m = V_m/(R_L + 1) \approx 67.7 \text{ mA}$ (66.3 mA with a 0.7 V diode drop). Peak inverse voltage of the diode: $\text{PIV} = V_m = 33.94 \text{ V}$.

With a real silicon diode ($V_F \approx 0.7 \text{ V}$) the peak becomes $V_m - V_F = 33.24 \text{ V}$, so $V_{dc} \approx 10.58 \text{ V}$, $V_{rms} \approx 16.62 \text{ V}$ (F.F., R.F. and η do not change).

3.3 Half-wave rectifier with R–C load

The capacitor charges to V_m at the peak and discharges through R_L while the diode is off. Since $\tau = 0.5 \text{ s} \gg T = 20 \text{ ms}$ the discharge is almost linear:

$$\begin{aligned}\Delta V_{pp} &\approx \frac{V_m}{f R_L C_L} = \frac{33.94}{50 \times 0.5} = 1.36 \text{ V}, & V_{dc} &\approx V_m - \frac{\Delta V_{pp}}{2} = 33.26 \text{ V}, \\ V_{ac} &\approx \frac{\Delta V_{pp}}{2\sqrt{3}} = 0.392 \text{ V (triangular ripple)}, & \text{R.F.} &= \frac{1}{2\sqrt{3} f R_L C_L} \approx 0.0118, \\ \text{F.F.} &= \sqrt{1 + \text{R.F.}^2} \approx 1.0001, & \eta &\approx 99.99\%.\end{aligned}$$

The diode conducts only from $\theta_1 = \arcsin(1 - \Delta V_{pp}/V_m) \approx 73.7^\circ$ to $\approx 90^\circ$ (about 16° per cycle), so the diode current is a short pulse of peak $i_{D,\max} \approx \omega C V_m \cos \theta_1 + V_m/R_L \approx 3 \text{ A}$ (in practice lower, limited by the transformer and wiring resistances). The diode now has to withstand $\text{PIV} = 2V_m \approx 67.9 \text{ V}$ (capacitor at $+V_m$, anode at $-V_m$).

3.4 Full-wave (bridge) rectifier with R load

D_1, D_2 conduct in the positive half cycle and D_3, D_4 in the negative one, so $v_L = V_m |\sin \omega t|$ with period $T/2 = 10 \text{ ms}$ (ripple frequency 100 Hz).

$$\begin{aligned}V_{dc} &= \frac{2V_m}{\pi} = 21.61 \text{ V}, & V_{rms} &= \frac{V_m}{\sqrt{2}} = 24.00 \text{ V}, & V_{ac} &= \sqrt{24^2 - 21.61^2} = 10.45 \text{ V}, \\ \text{F.F.} &= \frac{\pi}{2\sqrt{2}} = 1.111, & \text{R.F.} &= 0.483, & \eta &= \frac{8}{\pi^2} = 81.1\%.\end{aligned}$$

Two diodes are in series in the current path, so the peak is $V_m - 2V_F = 32.54 \text{ V}$, giving $V_{dc} \approx 20.72 \text{ V}$ and $V_{rms} \approx 23.01 \text{ V}$. PIV of each diode $= V_m$.

3.5 Full-wave (bridge) rectifier with R–C load

The capacitor is recharged twice per period, so the ripple is halved:

$$\begin{aligned}\Delta V_{pp} &\approx \frac{V_m}{2f R_L C_L} = 0.68 \text{ V}, & V_{dc} &\approx V_m - \frac{\Delta V_{pp}}{2} = 33.60 \text{ V}, & V_{ac} &\approx \frac{\Delta V_{pp}}{2\sqrt{3}} = 0.196 \text{ V}, \\ \text{R.F.} &= \frac{1}{4\sqrt{3} f R_L C_L} \approx 0.0058, & \text{F.F.} &\approx 1.0000, & \eta &\approx 100\%.\end{aligned}$$

Conduction angle $\approx 11.5^\circ$ per half cycle, current pulses $\approx 2.2 \text{ A}$ peak (theoretical), PIV of each diode $= V_m$.

3.6 Summary of the predicted values

Table 1: Theoretical values (ideal diodes / with $V_F = 0.7 \text{ V}$ per diode).

Circuit	Load	V_{dc} (V)	V_{ac} (V)	V_{rms} (V)	F.F.	R.F.	η
Half-wave	R	10.80 / 10.58	13.09 / 12.82	16.97 / 16.62	1.571	1.211	40.5 %
	R–C	33.26 / 32.58	0.392 / 0.384	33.26 / 32.58	1.000	0.012	99.99 %
Full-wave	R	21.61 / 20.72	10.45 / 10.02	24.00 / 23.01	1.111	0.483	81.1 %
	R–C	33.60 / 32.22	0.196 / 0.188	33.60 / 32.22	1.000	0.006	100 %

4 Circuits and procedure

4.1 Circuit diagrams

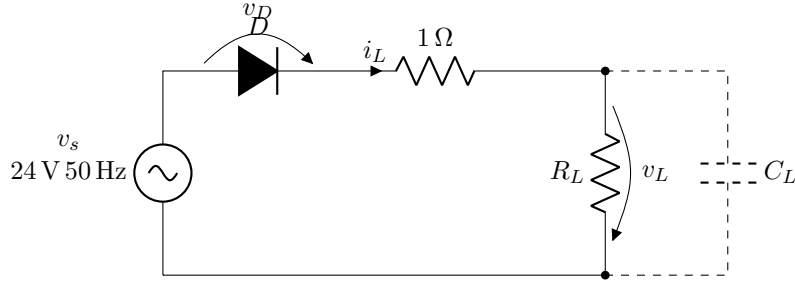


Figure 1: Half-wave rectifier. C_L (dashed) is connected only for the R–C load. The voltage across the $1\ \Omega$ resistor gives the current waveform.

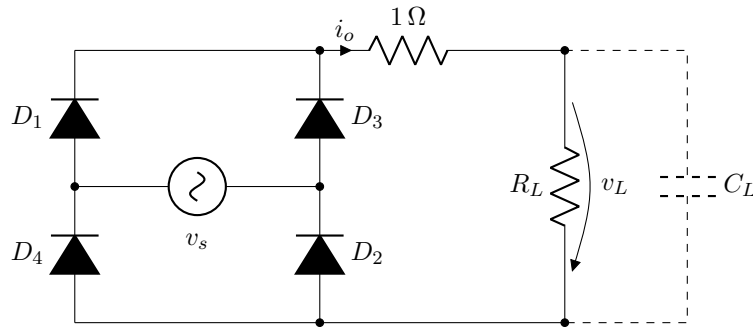


Figure 2: Full-wave bridge rectifier (D_1D_2 conduct for $v_s > 0$, D_3D_4 for $v_s < 0$).

4.2 Procedure followed

Part one – half-wave rectifier.

1. Switch on the supply and display the input voltage v_s (Fig. 3).
2. Connect D , the $1\ \Omega$ resistor and $R_L = 500\ \Omega$ as in Fig. 1.
3. Measure V_{dc} (“Mean”, DC coupling) and V_{ac} (“RMS”, AC coupling) of the load voltage; record in Table I.
4. Display the load voltage, the load current (voltage across $1\ \Omega$: $1\ \text{mV} \leftrightarrow 1\ \text{mA}$) and the diode voltage.
5. Switch off, add $C_L = 1000\ \mu\text{F}$ in parallel with R_L (observe the polarity of the electrolytic capacitor), switch on and repeat steps 3–4.

Part two – full-wave rectifier. Repeat the same steps with the bridge (D_1 – D_4) and record the results in Table II.

Precautions. Always switch off before rewiring; respect the capacitor polarity; in the bridge, never connect the two probe grounds to different points of the circuit (the common ground of the oscilloscope would short-circuit a diode) – display the input and the output one at a time.

Oscilloscope settings used: $5\ \text{ms/div}$ for all waveforms; $10\ \text{V/div}$ for voltages, $20\ \text{mV/div}$ for the current with R load, $1\ \text{V/div}$ for the current pulses with R–C load, $20\ \text{V/div}$ for the diode voltage with R–C load, $0.5\ \text{V/div}$ in AC coupling to zoom on the ripple.

5 Results

5.1 Measured values

Table (I) – Experimental results for the half-wave rectifier

Load	Value	V_{dc} (V)	V_{ac} (V)	$V_{rms} = \sqrt{V_{dc}^2 + V_{ac}^2}$ (V)	F.F.	R.F.	η
R	500 Ω	10.58	12.820	16.62	1.571	1.212	40.5 %
R-C	500 Ω , 1000 μ F	32.58	0.384	32.58	1.000	0.012	100.0 %

Table (II) – Experimental results for the full-wave rectifier

Load	Value	V_{dc} (V)	V_{ac} (V)	$V_{rms} = \sqrt{V_{dc}^2 + V_{ac}^2}$ (V)	F.F.	R.F.	η
R	500 Ω	20.72	10.020	23.02	1.111	0.484	81.0 %
R-C	500 Ω , 1000 μ F	32.22	0.188	32.22	1.000	0.006	100.0 %

Note: values shown are the *predicted* values. Replace them with your oscilloscope readings in the edit block at the top of the file.

Sample calculation (half-wave, R load): $V_{rms} = \sqrt{10.58^2 + 12.82^2} = 16.62$ V, F.F.= $16.62/10.58 = 1.571$, R.F.= $12.82/10.58 = 1.212$, $\eta = (10.58/16.62)^2 = 40.5$ %.

5.2 Comparison with theory

Table 2: Measured V_{dc} vs. theoretical (ideal-diode) V_{dc} .

Case	V_{dc} theory (V)	V_{dc} measured (V)	Error
Half-wave, R	10.80	10.58	2.0 %
Half-wave, R-C	33.26	32.58	2.0 %
Full-wave, R	21.61	20.72	4.1 %
Full-wave, R-C	33.60	32.22	4.1 %

5.3 Waveforms

All waveforms are drawn at 5 ms/div (2.5 periods on the screen); the vertical scale is given under each graph. They are drawn for ideal diodes; the measured waveforms are the same with peaks about 0.7 V (half-wave) or 1.4 V (bridge) lower.

Input voltage

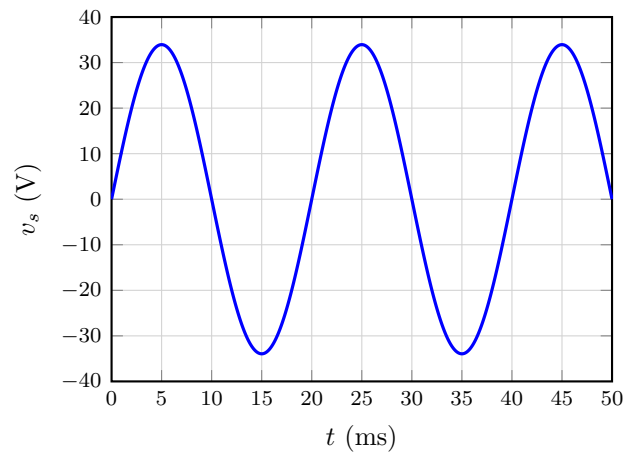
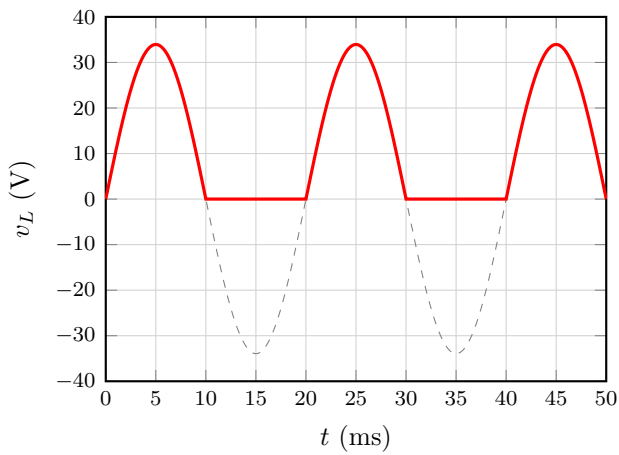
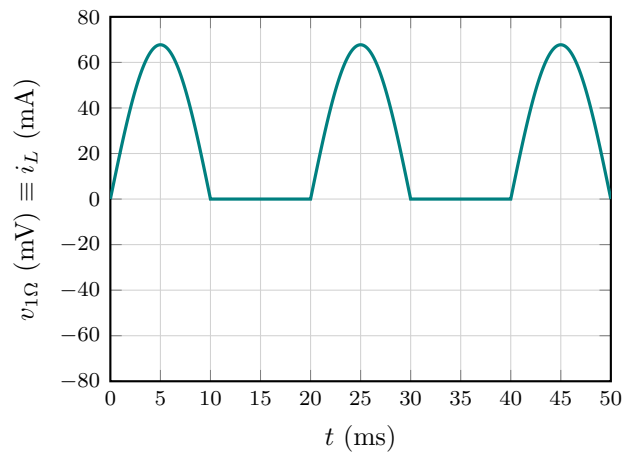


Figure 3: Input voltage: $V_m = 33.9 \text{ V}$, $T = 20 \text{ ms}$ (Time 5 ms/div, Volt 10 V/div).

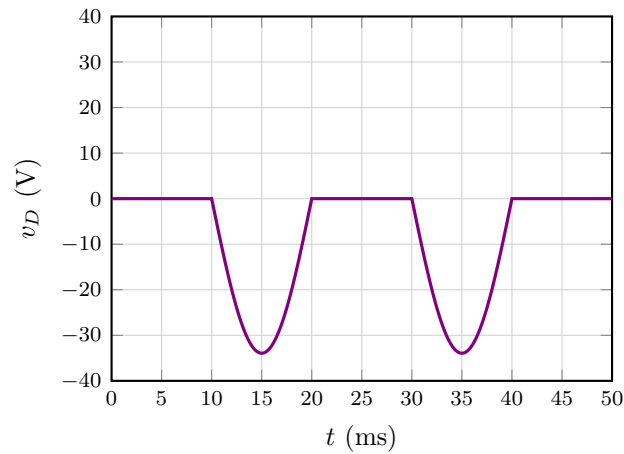
Half-wave rectifier – R load



(a) Load voltage (5 ms/div, 10 V/div)



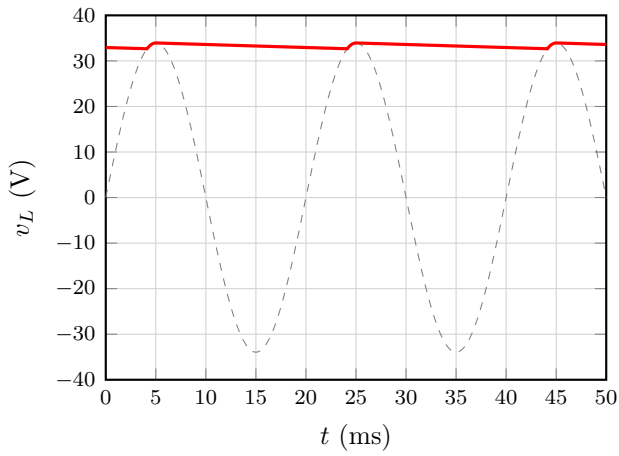
(b) Load current (5 ms/div, 20 mV/div)



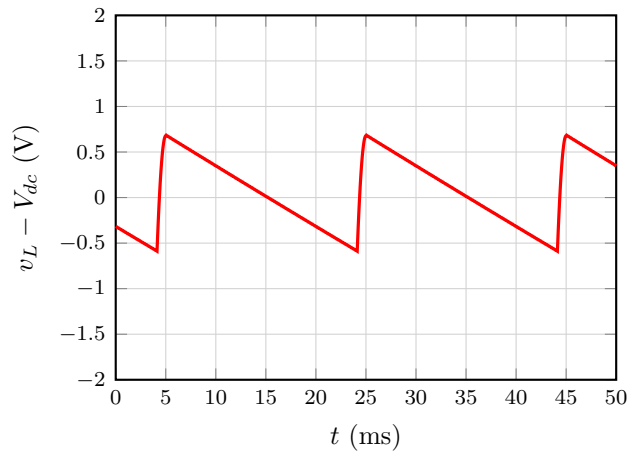
(c) Diode voltage (5 ms/div, 10 V/div)

Figure 4: Half-wave rectifier with R load.

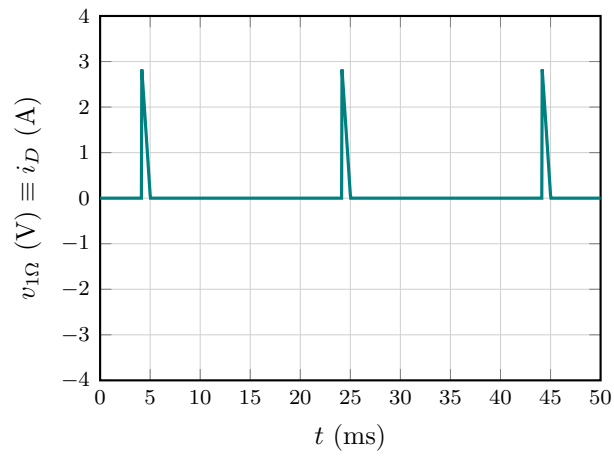
Half-wave rectifier – R–C load



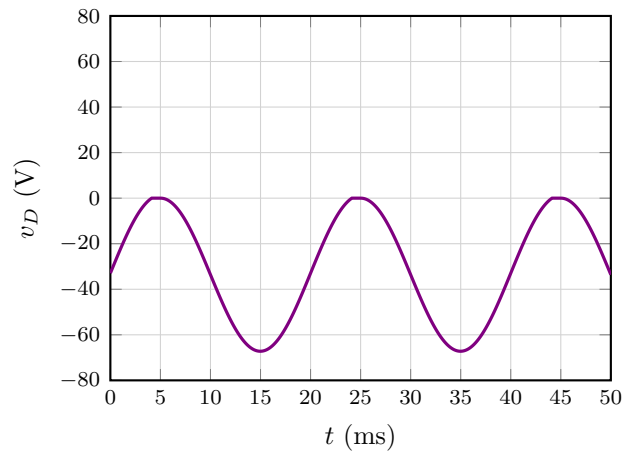
(a) Load voltage (5 ms/div, 10 V/div)



(b) Ripple, AC coupling (5 ms/div, 0.5 V/div)



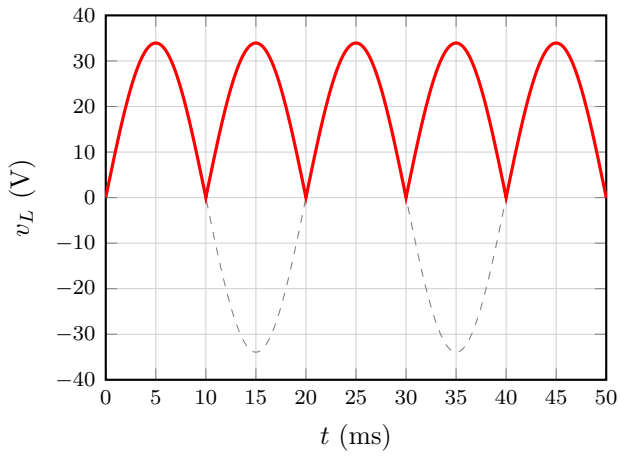
(c) Diode (charging) current (5 ms/div, 1 V/div)



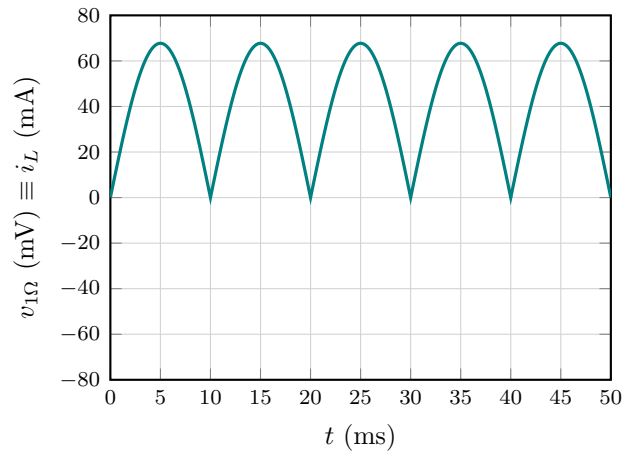
(d) Diode voltage, PIV $\approx 2V_m$ (5 ms/div, 20 V/div)

Figure 5: Half-wave rectifier with R–C load ($R_L = 500\,\Omega$, $C_L = 1000\,\mu\text{F}$).

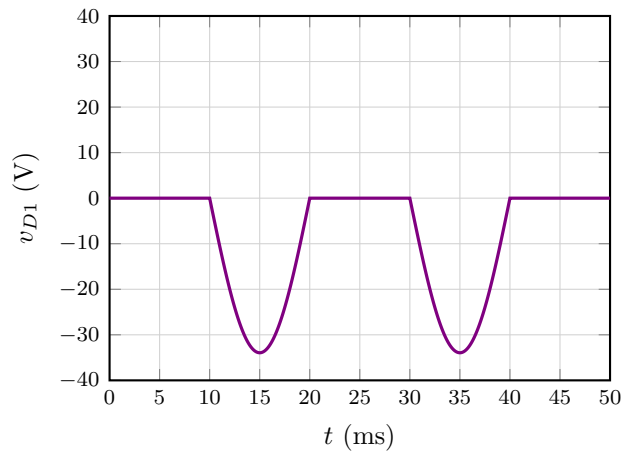
Full-wave rectifier – R load



(a) Load voltage (5 ms/div, 10 V/div)



(b) Load current (5 ms/div, 20 mV/div)



(c) Voltage across D_1 (5 ms/div, 10 V/div)

Figure 6: Full-wave bridge rectifier with R load.

Full-wave rectifier – R–C load

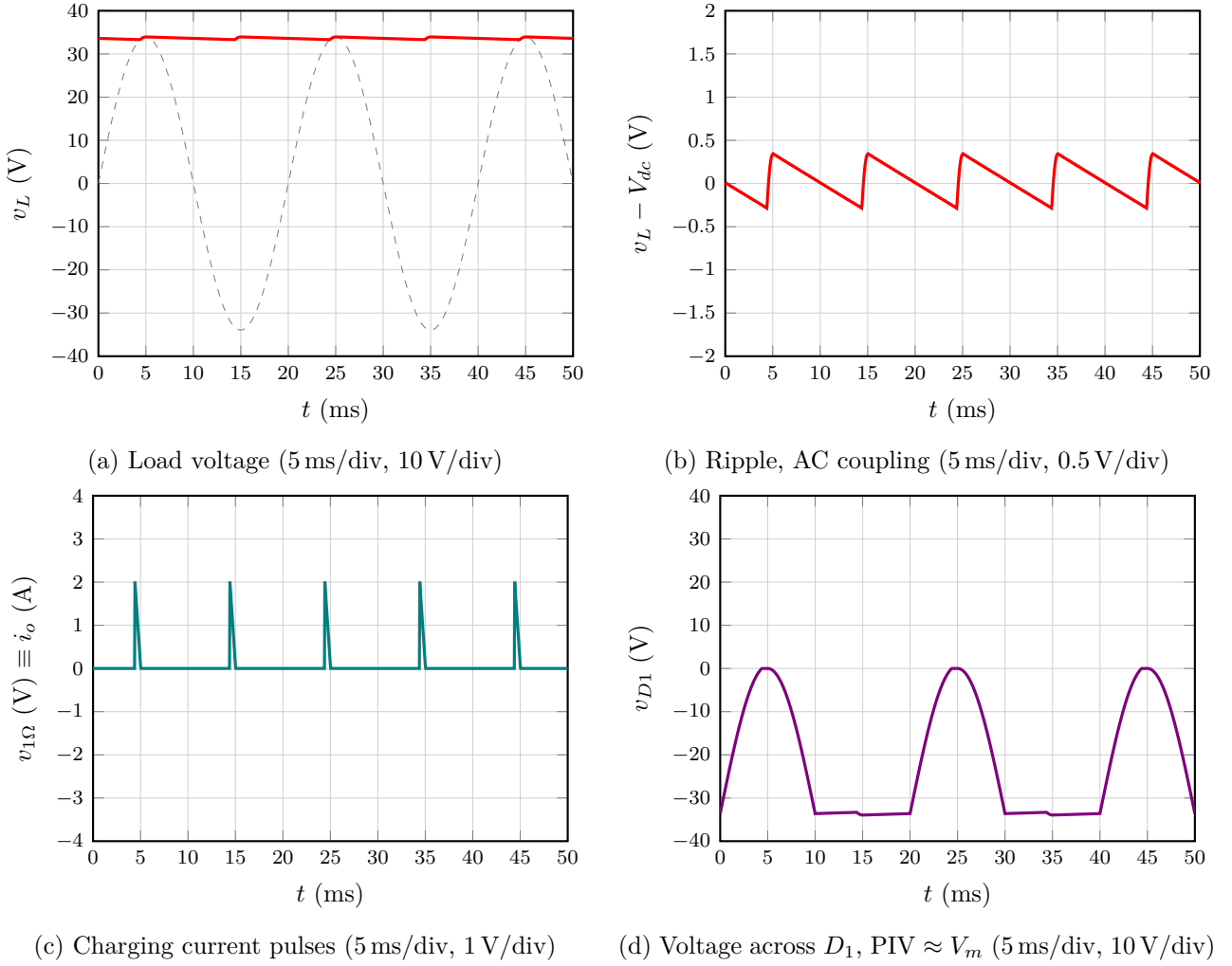


Figure 7: Full-wave bridge rectifier with R–C load.

6 Discussion and answers to questions

1. **Why is a $1\ \Omega$ resistor used to observe the current?** The oscilloscope only measures voltages. By Ohm's law $v_{1\Omega} = 1 \times i$, so the voltage across the $1\ \Omega$ resistor has exactly the shape (and numerically the value) of the current. Being very small compared with $R_L = 500\ \Omega$, it does not disturb the circuit.
2. **Half-wave vs. full-wave with R load.** The bridge doubles V_{dc} ($2V_m/\pi$ instead of V_m/π), reduces the ripple factor from 1.21 to 0.48, the form factor from 1.57 to 1.11, and doubles the efficiency ($40.5\% \rightarrow 81.1\%$). The ripple frequency is 100 Hz instead of 50 Hz, which is easier to filter. Its drawback is two diode drops in series instead of one.
3. **Effect of the capacitor.** The capacitor stores energy at the voltage peaks and feeds the load while the diodes are off; V_{dc} rises close to V_m and the ripple becomes very small ($\Delta V \approx V_m/(fRC)$ for half-wave, $V_m/(2fRC)$ for full-wave). F.F. $\rightarrow 1$, R.F. $\rightarrow$ about 1 % and $\eta \rightarrow 100\%$. A larger C or R_L (lighter load) gives less ripple. With the capacitor the bridge ripple is half of the half-wave ripple because the capacitor is recharged twice per period.
4. **Shape of the current with the capacitor.** The diodes conduct only during a short interval before each peak (about 16° half-wave, 11° full-wave), so the current is made of narrow, high pulses

(amperes) although the average load current is only $\approx V_{dc}/R_L \approx 65$ mA. These pulses stress the diodes and the transformer and distort the supply current.

5. **Diode voltage / PIV.** Half-wave R: $PIV = V_m \approx 34$ V. Half-wave R-C: $PIV = 2V_m \approx 68$ V (the capacitor keeps the cathode at $+V_m$ while the anode goes to $-V_m$), so the diode must be rated for twice the peak voltage. Bridge (R or R-C): $PIV = V_m$. When conducting, the diode voltage is only ≈ 0.7 V.
6. **Why do the measured values differ from the theory?** Forward voltage drop of the diodes (0.7 V, 1.4 V in the bridge); voltage drop in the transformer winding resistance (the supply voltage falls when loaded, mainly during the current pulses with the capacitor); the $1\ \Omega$ resistor and wiring resistance; the mains voltage is not exactly $24\ V_{rms}$ nor perfectly sinusoidal; tolerances of R_L and C_L (electrolytic capacitors $\pm 20\%$) and the ESR of the capacitor; accuracy and resolution of the oscilloscope measurements.
7. **Why not display the bridge input and output at the same time?** The two probe grounds are connected together (and to earth) inside the oscilloscope. In a bridge, the input terminals and the output terminals do not share a common point, so connecting both grounds would short-circuit a diode. Each waveform is displayed separately (or a differential probe is used).

7 Conclusion

In this experiment single-phase half-wave and full-wave (bridge) diode rectifiers were built and tested with R and R-C loads. With a pure resistive load the output is a pulsating DC voltage: the half-wave rectifier gives a low average value ($V_{dc} \approx V_m/\pi$), a large ripple (R.F. ≈ 1.21) and a poor efficiency ($\approx 40\%$), whereas the bridge rectifier doubles the DC value, uses both half cycles, produces a 100 Hz ripple (R.F. ≈ 0.48) and reaches $\approx 81\%$ efficiency. Adding the $1000\ \mu\text{F}$ filter capacitor raises the output close to the peak value (≈ 33 V) and reduces the ripple to around 1% (half-wave) and 0.6% (full-wave), the price being short, high charging current pulses in the diodes and, for the half-wave circuit, a diode reverse voltage of $2V_m$. The measured values agree with the theory, the small differences being explained mainly by the diode forward drops, the internal resistance of the source and the component tolerances. The full-wave bridge rectifier with capacitor filter is therefore the best of the four configurations for supplying a DC load.